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SUPERSEDING
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MILITARY SPECIFICATION

TELETYPEWRITER SETS AN/TGC-5 ( ) AND AN/TGC-5 ( )X

This specification has been approved by the Departments of the Army, the Navy, and the Air Force

1. SCOPE

1.1 This specification covers lightweight fixed station teletypewriter sets that operate from a power source of 115 volts a. c. ±10 percent at single phase frequencies of 50 to 60 cps. This equipment provides transmitting and receiving facilities for two full duplex polar or neutral circuits. Both equipments have the same makeup with the exception of the motors, Teletypewriter Set AN/TGC-5 ( ) contains a synchronous motor and Teletypewriter Set AN/TGC-5 ( ) X contains a series governed motor.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND DRAWINGS

2.1 The following specifications, standards, and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

NN-B-621—Boxes; Wood, Nailed and Lock-Corner.
NN-B-631—Boxes; Wood, Wirebound (For Domestic Shipment).
LLL-B-631—Boxes; Fiber, Corrugated (For Domestic Shipment).

MILITARY

JAN-P-104—Packaging and Packing for Overseas Shipment—Crates, Sheathed, Wood, Nailed.
JAN-P-105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN-P-106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
MIL-P-116—Preservation, Methods of.
JAN-T-152—Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment: General Process for.
MIL-T-10513—Tropicalization of Materials Used in Signal Corps Equipment.
MIL-P-11268—Parts, Materials and Processes Used in Signal Corps Equipment.
MIL-S-11748—Suppression Requirements for Electrical and Electronic Equipment (Except Internal Combustion Engine-Driven Equipments).
MIL-R-11985—Reperforator — Teletypewriter TT-107( )/FG, TT-108( )/FG, TT-109( )/FG and TT-110( )/FG.
MIL-E-15090—Enamel, Equipment, Light-Gray (Formula No. 111).

STANDARDS

MILITARY

JAN-STD-15—Electrical and Electronic Symbols.
MIL-STD-105—Sampling Procedures and Tables For Inspection by Attributes.
MIL-STD-129—Making of Shipments.
MIL-STD-130—Identification Marking of U. S. Military Property.

DRAWINGS

SIGNAL CORPS

460-12—Group B-2 Sampling Plans.
SC-D-15914—Moisture Resistance Test Cycle for Ground Signal Equipment.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions may be obtained from or as

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directed by the contracting officer. Both title and identifying number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 Description.—Teletypewriter Set AN/TGC-5 ( ) shall consist of a transmitting console and a receiving and monitoring console. The transmitting console shall contain a dual tape transmitter and a switching control unit. The receiving and monitoring console shall contain a d. c. rectifier and two typing reperforators one of which is normally used for receiving and the other for monitoring. Synchronous motors shall be provided. Teletypewriter Set AN/TGC-5 ( ) X shall be identical to Teletypewriter Set AN/TGC-5 ( ) except that series governed motors shall be provided.

3.2 Preproduction samples.

3.2.1 *Quantities.*—After award of contract, in addition to the samples listed in Specification MIL-P-11268, two complete equipments shall be furnished for performance testing.

3.2.2 *Compliance with specified requirements.*

3.2.2.1 Preproduction samples shall meet specified requirements, shall be made and assembled by tools and methods that will be used for quantity production, and shall be accompanied by a statement to that effect (see 3.2.2.3 for exception).

3.2.2.2 Approval of preproduction samples shall not be construed as a waiver of any specified requirement. After being released to the contractor (see 3.2.5), preproduction samples to be offered as units on contract shall be refabricated by the contractor if necessary to meet specified requirements.

3.2.2.3 When deviation from 3.2.2.1 is unavoidable, the preproduction samples may be submitted for approval provided that the accompanying statement describes in detail each nonconforming feature, reason therefor, and manner in which it will be corrected in production of equipment on contract.

3.2.3 *Test data.*—The preproduction samples shall be accompanied by test data showing compliance with specified performance. The test data shall comprise an engineering report giving test procedures, observations, and other data, calculations, test results, and essential details of the testing equipment (manufacturer's model, serial number, date of calibration, and the like).

3.2.4 *Preproduction inspection.*—Preproduction samples may be subjected by the Government to the group A inspection and nondestructive group B and group C inspection specified in tables II, III, and IV, respectively, and to the preproduction inspection specified in table I. Unlisted nondestructive inspection on preprodution samples of equipment and unlisted destructive or nondestructive inspection on separate preproduction samples of parts or materials may be performed to determine compliance with specified requirements. Preproduction inspection will normally be performed in this order: (1) vibration and (2) bounce; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Table I.—Preproduction Inspection

Inspection (For additional preproduction inspection see 3.2.4.)	Rqt. par.	Test par.
Service conditions	3.4	4.10

3.2.5 *Reference standards.*—After preproduction samples have been approved and returned to the contractor's plant they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Parts, materials, and processes; general.—Parts, materials, and processes shall conform to Specification MIL-P-11268, unless otherwise specified herein.

3.4 Service conditions.—The equipment shall meet the following service conditions:

3.4.1 *Equipment operating.*—The equipment shall meet the requirements of this specification under any probable combination of the following:

(a) *Operation.*

(1) 300 hours at 368.1 operation per minute (o. p. m.) or 150 hours at 600 o. p. m. before necessity of minor mechanical adjustment, with exception of range orientation, motor syn-

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chronization, and minor lubrication.

(2) 1,000 hours at 368.1 o. p. m. or 500 hours at 600 o. p. m. before requiring mechanical adjustment.

(3) 4,000 hours at 368.1 o. p. m. or 2,000 hours at 600 o. p. m. before mechanical degradation beyond permissible tolerances, would require replacement of any part or assembly.

(b) *Temperature.*—Any ambient temperature in the range of +32 to +132° F., including sun load.

(c) *Elevation.*—Any elevation up to 10,000 feet above sea level.

(d) *Relative humidity.*—Any relative humidity up to 95 percent.

3.4.2 *Equipment nonoperating.*—The equipment shall meet the requirements of 3.4.1 after subjection to any one or more of the following nonoperating conditions:

(a) *Temperature.*—Continuous exposure for 72 hours at 160° F. and 72 hours at −80° F., at any orientation.

(b) *Elevation.*—Any elevation up to 50,000 feet above sea level.

(c) *Relative humidity.*—Relative humidity up to 100 percent including condensation caused by temperature changes. For 16 hours after subjection to these conditions, the equipment shall be allowed to return to normal room temperature and operation as specified herein.

(d) *Orientation.*—Indefinite storage at any orientation.

(e) *Vibration and bounce.*—As required in 4.10.5 and 4.10.6.

3.5 Adaptability to alterations.—The minimum practicable variety of parts shall be used throughout the equipment and shall be such that any part, within the limitating tolerances, characteristics, rating, etc., prescribed by the specifications for the part concerned, may be used to replace the corresponding part in the equipment. It is intended that the equipment shall utilize, as far as possible, the subassemblies and parts common to the lightweight Reperforator-Teletypewriter Set AN/GGC-3 ( ). This will provide greater interchangeability of parts between sets and will alleviate the problem of stock and supply of teletypewriter replacement parts.

3.6 Operation with associated equipments.—Operation shall be on the standard 7.42-unit start-stop Baudot code, permitting interoperation with all existing military teletypewriter equipment and similar equipment under development. In addition, provision shall be made to operate with synchronous "on-line" security equipment now under development.

3.7 Field maintenance.—Field maintenance on this equipment shall be capable of being performed with existing standard military test equipment.

3.8 Equipment mounting.—All equipment units shall be mounted and connected in such manner that they may be completely removed by unclamping the mechanical mountings and unplugging any electrical connections.

3.9 Cabinet wiring.—There shall be no cabinet wiring as such. All interconnecting cables shall be fitted with plugs that may be plugged into receptacles on the units to which they must be connected. One end of all cables shall be mechanically fixed to some component, in order that there shall be no loose cables. Means shall be provided for securing the cables neatly to the sides of the consoles. Plugs and sockets for d. c. wiring shall be midget twist lock type with a center pin used as a ground connection between units. Plugs and sockets for a. c. power shall be standard parallel blade types. Plugs and jacks for the relay control unit shall be multiple connectors with grounding wires included.

3.10 Detail requirements.

3.10.1 *Equipment arrangements.*

3.10.1.1 *As a single unit.*—The sending and receiving consoles are installed side by side and interconnections made by means of short cables. The following table gives the types of signaling required for the various modes of operation.

	Transmitting	*Receiving*
Normal Single	20 or 60 Ma Neutral	20 or 60 Ma Neutral.
Normal Duplex	20 or 60 Ma Neutral or 30 Ma Polar	20 or 60 Ma Neutral or 30 Ma Polar.
Split Single	20 or 60 Ma Neutral	20 or 60 Ma Neutral.
Split Duplex	20 or 60 Ma Neutral or 30 Ma Polar	20 or 60 Ma Neutral or 30 Ma Polar.

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3.10.1.2 *As a torn-tape relay system.*—The transmitting consoles of a group of sets are mounted in a line. Half of the receiving consoles, equipped with monitor reperforators taken from the remaining half, are arranged in a line approximately 4 feet from the transmitting consoles. The remaining half of the receiving reperforators are arranged in a line approximately 4 feet from the transmitting consoles but on the opposite side from the monitor consoles. Interconnection between the units is made by means of long cables. The operators of the transmitting consoles are also responsible for rerunning tapes taken from the monitor reels. The following table gives the types of signaling required for the various modes of operation.

	Transmitting	*Receiving*
Normal Single	20 or 60 Ma Neutral	20 or 60 Ma Neutral.
Normal Duplex	20 or 60 Ma Neutral or 30 Ma Polar	20 or 60 Ma Neutral or 30 Ma Polar.

3.10.1.3 *As a multiple address processing unit.*—It shall be possible to connect, by means of patch cords, several receiving consoles to one transmitting console. By this means, several message tapes may be made simultaneously. It shall also be possible to patch a page printer into the send circuit so that keyboard operation to each or all of the receiving consoles is possible.

3.10.2 *Physical characteristics.*

3.10.2.1 *Weight.*—The combined total weight of the sending and receiving consoles shall not exceed 275 pounds when set up for operation.

3.10.2.2 *Height.*—The maximum height of any console with equipment mounted thereon shall not exceed 60 inches.

3.10.2.3 *Floor space.*—The maximum floor space required by the sending and receiving consoles together shall not exceed 4 square feet.

3.10.3 *Operating speeds.*—It shall be possible to adjust the speeds of all teletypewriter apparatus to either 368, 404, 460, or 600 operations per minute by appropriate gear changes.

3.10.4 *Power supply.*

3.10.4.1 *Teletypewriter Set AN/TGC–5( )X.*—This equipment shall be designed to operate from a 115-volt, plus or minus 10 percent; 50/60 cycle (c. p. s.) power source. Current shall not exceed 10 amperes.

3.10.4.2 *Teletypewriter Set AN/TGC–5( ).*—This equipment shall be designed to operate from a 115-volt, plus or minus 10 percent; 60-cycle±3 c. p. s. power source. Current shall not exceed 10 amperes.

3.10.5 *Transmitting console.*

3.10.5.1 *Description.*—This console shall consist of a lightweight aluminum cabinet on which is mounted a dual tape transmitter. The height of the console is 37 inches which from the floor is identical to that at which the transmitters are mounted in Teletypewriter Set AN/TGC–1. In this cabinet shall be mounted an indicator panel and a jack mounted relay control unit.

3.10.5.2 *Dual tape transmitter.*

3.10.5.2.1 *Motor types.*—Teletypewriter Set AN/TGC–5( ) shall be provided with a dual tape transmitter driven by a synchronous motor of proper capacity for continuous operation. Teletypewriter Set AN/TGC–5( )X differs only in that a series-governed motor of proper capacity for continuous operation shall be provided. Power circuits shall be adequately fused.

3.10.5.2.2 *Motor adjustment.*—Each series-governed motor shall be equipped with a governor designed to maintain proper speed at each of the operating speeds of 3.10.3 with a tolerance of plus or minus 0.5 percent. A three-spot target shall be mounted on the armature shaft for checking motor speed using a tuning fork with a vibration rate of 180 vibrations per second.

3.10.5.2.3 *Operation as a tape reader.*—Each tape transmitter shall be capable of reading perforated tape on a five-wire basis. During this mode of operation, the transmitter start magnet shall be operated on a "pulse per operation" basis by pulses furnished by external equipment. The intelligence read from the tape shall be available immediately after receipt of the external pulse and shall remain so for approximately one half of the operation length. During the remainder of the cycle, the five-wire circuits shall be open. The start magnet control lever of 3.10.5.2.4 shall be in the upper position for this test.

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3.10.5.2.4 *Operation as a sequential tape transmitter.*—Each tape transmitter shall be capable of normal operation from either 11/16 inch or 7/8 inch wide tape fully perforated or chadless. During this mode of operation, the start magnet is energized by a steady current during transmission and not energized when at rest. The start magnet electrical circuit shall be controllable by means of a lever. This lever shall have three positions. In the bottom position, the start magnet shall not be energized and the means of tape feed shall be disengaged from the feed holes in the tape. In the middle position, the start magnet shall not be energized and the means of tape feed shall engage the feed holes in the tape. In the upper position, the start magnet shall be energized and the means of tape feed shall feed the tape through the tape guide.

3.10.5.2.5 *Operation as a markable tape transmitter.*—Each transmitter shall be capable of operation as a markable tape transmitter to perform the function of numbering messages. The intelligence shall be coded on rotary switches which shall step under control of the tape transmitter.

3.10.5.2.6 *Generation of blank signals.*—Each transmitter shall be capable of sending blank signals to the line when the "end of tape" sensing mechanism has sensed the end of a message tape. The number of blank characters shall be adjustable from ten to twenty and shall be capable of manual disablement. The electrical contacts operated by the end-of-tape sensing mechanism shall be operated at the end of the blank signals when the transmitter is adjusted for this mode of operation.

3.10.5.2.7 *General.*—Each transmitter shall be capable of transmitting from both tapes and tape loops. The tape shall be properly alined with respect to the sensing pins during operation, and tape feed shall be uniform and positive. An end-of-tape sensing mechanism shall be furnished which shall cause contacts to open when the end of the tape has cleared the sensing pins.

3.10.5.3 *Relay control unit.*—A relay control unit shall be mounted within the transmitting console which shall permit all of the modes of operation as outlined in this specification. On this unit shall be mounted the necessary switches to select the various modes of operation.

3.10.5.4 *Cabinet.*—The transmitting console cabinet shall be as small and as light weight as possible consistent with the components to be mounted within. Tape bins shall be provided in the rear of the cabinet for each transmitter to receive the spent tapes. A removable tape bin shall be provided at the front of the cabinet to hold the ends of message tapes that have been inserted in the transmitters for sending. A tape holder grid shall be provided on top of the transmitting console. It shall provide 4 columns of slots vertically, 28 slots per column, for holding standard perforated tapes (both 11/16 and 7/8 inches in width) and shall be readily accessible from the rear for tape insertion, and from the front for tape removal. Designation card holders shall be provided for each column on the tape holder grid.

3.10.5.5 *Indicator panel.*—This panel shall be in duplicate, each part serving one line while in split operation. It shall be mounted on the top of the transmitting console for easy access by an operator. In normal operation, one part shall be in use and the other shall be available as a spare unit. On this panel shall be mounted a "break" indicating lamp with associated control button, and a "message received" lamp with associated control button.

3.10.6 *Receiving console.*—This console shall consist of a light weight aluminum cabinet in which shall be mounted a rectifier, a monitor typing reperforator, and a receiving typing reperforator.

3.10.6.1 *Rectifier.*—This rectifier shall be of the dry-disc type of sufficient capacity to supply all of the d. c. requirements of the teletypewriter set. Nominal d. c. output voltage shall be 120 volts, and taps shall be provided to compensate for an input power of 115±10 percent volts and 50/60 cycles and for aging of the rectifying elements. Ripple voltage shall not exceed 0.5 volts. Four a. c. outlets shall be provided on the unit so that plugs from the dual transmitter, the receiving typing reperforator, and the monitoring typing reperforator may be inserted to obtain a. c. power for operation. Both sides of input power circuits shall be adequately fused as well as live side of d. c. power. Outlets for

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d. c. shall be provided commensurate with the requirements of this specification. Voltage regulation from minimum load to maximum load shall not exceed 10 percent.

3.10.6.2 *Monitor typing reperforator.*—This unit shall be a Reperforator Teletypewriter TT-107/FG in Teletypewriter Set AN/TGC-5( ) and a Reperforator Teletypewriter TT-109/FG in Teletypewriter Set AN/TGC-5( )X equipped with a tape winder. Power to operate the tape winder shall be derived from the motor of the typing reperforator. A tape deflector shall be provided that will deflect the message tape from the typing reperforator through an opening in the front of the receiving console when desired. Reinking of the teletypewriter ribbon shall be provided.

3.10.6.3 *Receiving typing reperforator.*—This unit shall be Reperforator Teletypewriter TT-107/FG in Teletypewriter Set AN/TGC-5( ) and a Reperforator Teletypewriter TT-109/FG in Teletypewriter Set AN/TGC-5( )X. A tape deflector shall be provided that will deflect the message tape from the typing reperforator through an opening in front of the receiving console. Reinking of the teletypewriter ribbon shall be provided.

3.10.6.4 *Cabinet.*—The receiving cabinet shall be as small and as light weight as possible consistent with the components to be mounted within. Tape bins shall be mounted on the front of the cabinet to receive the ends of the messages as they issue from the typing reperforators. Both typing reperforators shall be mounted on captive slide drawers that shall be capable of sliding out either the front or rear of the console for paper tape changing and inspection. Designation card holders shall be provided for each reperforator position. Message number card holders shall be provided on the front of the console adjacent to each reperforator position. The number card shall be 3¾ inches square and shall be held at a convenient angle to permit the operator to make notations thereon.

3.10.7 *Special features.*

3.10.7.1 *Break-lock.*—A "break-lock" feature shall be provided to prevent transmission into an open line. It shall be effective when the teletypewriter set is conditioned for operation on one or two full-duplex (polar or neutral) or half-duplex (neutral) 20 or 60 milliampere d. c. lines as follows:

3.10.7.1.1 *Normal operations.*—When the teletypewriter set is conditioned for operation on one full-duplex (neutral 20 or 60 MA or polar) or half-duplex (neutral 20 or 60 MA) transmitting line, and the line reflects a prolonged break or open, the transmitter shall stop sending, the transmitter and monitor reperforator motors shall be turned off, and a break signal indicator lamp (red) shall light. The transmitter shall remain locked in this state, regardless of the line condition. Manual unlocking of the transmitter and extinguishing of the break-lamp shall be accomplished by operating a push-button release key when the "open" has been removed from the line.

3.10.7.1.2 *Split operation.*—When the teletypewriter set is conditioned for operation on two full-duplex (neutral 20 or 60 MA or polar) or half-duplex (neutral 20 or 60 MA) transmitting lines with the transmitters split and sending to both lines, a prolonged break on one line only, shall not stop the transmitter motor, but shall stop transmission on that line, and light the associated break signal indicator lamp. A prolonged break on both lines shall stop the transmitter motor, stop transmissions to both lines and light both break signal indicator lamps. When the transmitter is sending to one of the two lines only, a prolonged break on this line shall stop the transmitter motor, stop transmission to the line, and light the associated break signal indicator lamp. It shall, however, be possible to send on the remaining line if it is not open. During split operation, the monitor reperforator shall be used as a receiver and its motor shall run continuously.

3.10.7.2 *Motor control time delay.*—A "motor control time delay" feature shall be provided which will stop the motors of the monitor reperforator and transmitter when the transmitters stop sending. It shall also provide a delay in start of transmission to allow the motors to attain operating speed.

3.10.7.3 *Alternate transmitter operation.*—When operating normal full-duplex (neutral 20 or 60 MA or polar) or half-duplex (neutral

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20 or 60 MA), the message transmitters shall automatically operate alternately.

3.10.7.4 *Reperforator lock-out.*—A reperforator lock-out feature shall be provided for half-duplex operation. When one half-duplex line is used, with a receiving reperforator and a monitor reperforator, in series, the receiving reperforator shall be locked-out during local transmission to the line. During distant transmission to the line, the monitor reperforator shall be locked-out and the receiving reperforator shall receive. When two separate half-duplex lines are in use, both reperforators shall be used as receivers only and shall be locked-out during local transmission to their respective lines.

3.10.7.5 *Local test circuit.*—An internal local test circuit shall be provided to enable the operator to check the operation of the equipment off line. This circuit shall be obtained by manual switching and shall connect the dual tape transmitter and reperforators to the rectifier on either a 20 or 60 milliampere neutral or 30 milliampere polar loop basis within the teletypewriter set. If equipment is furnishing line current, it shall continue to do so. If it is not, lines shall be shorted.

3.10.7.6 *Time-date stamping.*—Provision shall be made to stamp the time on all reperforator tapes at one-minute intervals if the tape has moved since the previous stamping. Letters shall be of sufficient size to be easily readable when stamped over fully perforated tape. Time stamping may be by means of a standard self-stepping stamp actuated by internal control or from a master clock giving one-minute pulses.

3.10.7.7 *Automatic numbering.*—Automatic numbering shall be provided, capable of manual forward stepping plus push-button reset to number 1 for messages transmitted. It shall be made for manual disablement of the automatic numbering. The exact sequences to be sent out are listed below.

Case 1. (5 spaces) (2CR) (LF) (Channel Designation) (Channel Letter) (FIG) (Channel Number) (LTRS)

Case 2. (LTRS) ZCZC (Channel Designation) (Channel Letter) (FIG) (Channel Number) (LTRS)

It shall be possible to change from one case to another by a switch adjustment. The channel designation and channel letter shall be changeable on a plug and terminal basis by patching the proper five-wire intelligence.

3.10.7.8 *Indicator panel operation.*—The indicator panel described in 3.10.5.5 shall operate as follows:

3.10.7.8.1 *Break indication.*—When a prolonged break in the transmitting line occurs, the break lamp associated with that line shall light. It shall remain lighted to inform the operator of the line break. The operator may press the associated control button to extinguish the break lamp when the line break has been removed.

3.10.7.8.2 *Message received indication.*—When a message is received on a receiving line, the "message received" lamp, associated with that line, shall light, if the associated control switch has been positioned by the receiving operator. The lamp shall remain lighted to inform the operator that a message is being received. The operator may operate the control switch to extinguish the "message received" lamp.

3.10.7.9 *Jack facilities.*—Jack facilities shall be provided in each line circuit being used by the teletypewriter set.

3.10.7.10 *Line terminations.*

3.10.7.10.1 *Receiving console.*—The receiving console shall provide line terminations for connecting two full-duplex receiving circuits, when the receiving console is operating alone as either a receiving console or a monitoring console. The line terminals shall be located at the rear of the console and shall be readily accessible for line connection.

3.10.7.10.2 *Transmitting console.*—The transmitting console shall provide for connection from one or two full-duplex or half-duplex signal lines. The line terminals shall be located at the rear of the console and shall be readily accessible for line connection.

3.11 Equipment finish.—The color of the equipment finish shall conform to Specification MIL-E-15090.

3.12 Circuit diagrams.—Circuit diagrams (schematic, wiring and cabling) shall be furnished as required to facilitate installation, operation and maintenance. The diagrams shall

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be in the form of labels meeting the general requirements of Specification MIL-P-11268.

3.12.1 *Schematic and wiring diagrams.*—Schematic and wiring diagrams shall be suitable for use by maintenance personnel, and shall be mounted so far as practical on inside surfaces where they will not be exposed to abrasion or the elements but will be readily accessible to maintenance personnel. Graphical symbols on schematic diagrams shall conform to Standard JAN-STD-15.

3.12.2 *Facsimiles.*—In addition to any facsimiles specified by applicable documents, 4 copies of the circuit diagrams shall be submitted for approval to the contracting officer (see 6.2).

3.13 Marking.—Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Radio interference suppression.—The equipment shall conform to the requirements of Specification MIL-S-11748 and the interference limits set forth therein, with the antenna of the interference measuring equipment located 2 feet from the equipment under test in order to eliminate interference to high sensitivity radio receiving equipment.

3.15 Interchangeability.—Corresponding components on contract shall be physically and functionally interchangeable as complete units. Components shall be so constructed that any subassembly or part meeting specified dimensions, ratings, characteristics, etc. can be substituted for a corresponding, replaceable subassembly or part without modification of the substituted unit or of the equipment and without causing the equipment to deviate from specified requirements. When dimensions, ratings, characteristics, etc. are not specified, the manufacturer's design limits shall be used to determine compliance with the foregoing. Whether subassemblies and parts are replaceable shall be as determined by the contracting officer.

3.16 Tropicalization.—The assembled equipment shall be treated in accordance with Specification JAN-T-152. Before being assembled into the equipment, materials and parts shall be treated in accordance with Specification MIL-T-10513.

3.17 Technical literature, tools, and running spare parts.—Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2.)

3.18 Dielectric strength.—All circuits shall meet the dielectric strength test specified in 4.4.

3.19 Bounce conditioning.—After final assembly, each component on contract, with shock mounts removed or blocked, shall be placed in its normal operating position on the table of the Package Tester as made by the L. A. B. Corporation, Summit, N. J., or equal. The Package Tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$ inch thick strip of material under one corner or edge of the equipment to a distance of 3 inches as the equipment bounces. After conditioning, the equipment shall be capable of meeting the inspection of section 4.

3.20 Workmanship.—Components, subassemblies, and parts shall be manufactured and assembled in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection, general.—Government inspection shall consist of group A, group B, and group C inspection as specified in 4.1.1 through 4.1.3.2. (See 6.3.1.) Group B inspection shall normally be performed on lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. In addition, group C inspection shall normally be performed on sample units that have been subjected to and met group B inspection. However, the order may be varied when the Government considers it more practical to select separate samples for group B or group C inspection, or both.

4.1.1 *Group A inspection.*—Group A inspection shall be in accordance with table II and Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government inspector, except that the operational inspection (4.9) shall be last.

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Table II.—Government Group A Inspection

Inspection or test	Req. par.	Test par.	AQL	
			Major	Minor
Visual and Mechanical (including inspection of all equipment except typing reperforators which are covered by Specification MIL-R-11985.	----------	4. 3	10 dphu------	40 dphu.
Electrical and Systems Tests (60 Speed):*				
Dielectric----------	3.18	4.4		
Continuity----------	----------	4.5		
Motor Adjustment—60 speed (AN/TGC-5( )X only).	3.10.5.2.2	4.6.1		
Operation as a tape reader—60 speed----------	3.10.5.2.3	4.6.2		
Input test----------	3.10.6.1	4.7.1		
Regulation----------	3.10.6.1	4.7.2		
Ripple Voltage----------	3.10.6.1	4.7.3		
Break lock-normal single operation----------	3.10.7.1.1	4.9.1	10 dphu for the group.	
Break lock-normal duplex operation----------	3.10.7.1.1	4.9.1.1		
Break lock-split single operation----------	3.10.7.1.2	4.9.2		
Break lock-split duplex operation----------	3.10.7.1.2	4.9.2.1		
Motor Control Time Delay----------	3.10.7.2	4.9.3		
Alternate Transmitter Operation—60 speed---------	3.10.7.3	4.9.4		
Reperforator Lock-Out-Normal Operation----------	3.10.7.4	4.9.5		
Reperforator Lock-Out-Split Operation----------	3.10.7.4	4.9.5.1		
Generation of Blank Signal Test—60 speed---------	3.10.5.2.6	4.9.6		
Local Test Circuit Test—60 speed----------	3.10.7.5	4.9.7		
Automatic numbering test—60 speed----------	3.10.7.7	4.9.8		

*All electrical defects will be classified as major.

4.1.2 *Group B inspection.*—Group B inspection shall be in accordance with table III.

Table III.—Government Group B Inspection

Inspection or test	Req. par.	Test par.	Sampling plan	
			Drawing	AQL
Systems Tests (100 speed):				
Motor Adjustment—100 speed (AN/TGC-5( )X only)-------	3. 10. 5. 2. 2	4. 6. 1. 1	460-12	5%
Operation as a tape reader—100 speed----------	3. 10. 5. 2. 3	4. 6. 2 1.		
Alternate Transmitter Operation—100 speed----------	3. 10. 7. 3	4. 9. 4. 1		
Generation of Blank Signal Test—100 speed----------	3. 10. 5. 2. 6	4. 9. 6. 1		
Local Test Circuit Test—100 speed----------	3. 10. 7. 5	4. 9. 7. 1		
Automatic numbering test—100 speed----------	3. 10. 7. 7	4. 9. 8. 4		

4.1.2.1. *Order of inspection within group B.*—Group B inspection shall be performed in any order which is satisfactory to the Government inspector.

4.1.2.2 *Sampling procedure.*—Samples shall be selected in an unbiased manner by the Government inspector. Unless otherwise specified, inspection shall be started using level II of the specified drawing. If the number of defective sample units from a lot exceeds the acceptance number specified on the drawing, the lot shall be rejected. Disposition of rejected product (sample units and lots) shall be in accordance with Standard MIL-STD-105 and 4.1.5.

4.1.2.3 *Procedure in case of failure.*—When a lot is rejected, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results

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thereof and proposed corrective action. If the contractor and Government inspector cannot agree on the corrective action to be taken, the matter shall be referred to the contracting officer for resolution.

Table IV.—Government Group C Inspection

Inspection or test	Req. par.	Test par.
Radio Interference Suppression	3. 14	4. 9. 9

4.1.3.1 *Selection of samples.*—One equipment for group C inspection shall be selected in an unbiased manner by the Government inspector from either each months production or from each group of 50 units if the months production is 50 units or more.

4.1.3.2 *Noncompliance.*—If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results thereof and proposed corrective action to be taken on the process and affected portion of product. If the Government inspector does not consider that the corrective action will enable the equipment to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

4.1.4 *Reinspection to conforming group B and group C sample units.*—Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection (after repair of all visible damage).

4.1.5 *Disposition of nonconforming product.*—When defective sample units or rejected lots are resubmitted or affected portion of product is submitted for acceptance, such items shall be suitable tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The record shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.2 Test on preproduction samples.—The preproduction samples shall be tested to meet the requirements of 3.2.4.

4.3 Visual and mechanical inspection.—Parts and equipment shall be inspected for workmanship, mechanical fit, loose nuts and screws, application of specified tropicalization treatments, and miscellaneous defects. Controls and fastening devices shall be inspected for mechanical operation. Wiring, soldered connections, ground connections, welds, finished, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of Standard MIL-STD-105.

4.4 **Dielectric test.**—When practicable and provided no unavoidable unsoldering of connections is required and components are not overrated by the applied voltage, the dielectric strength of all circuits on the equipment shall be tested to withstand 400 volts or two times the maximum peak operation voltage, whichever is greater. The test voltage shall be d. c. or peak a. c. of the same value, and shall be applied for 10 seconds between wiring and adjacent nonconductors and ground.

4.5 **Continuity test.**—Electrical continuity of wiring and other connections, including ground connections to the chassis, shall be verified by a point-to-point test except when such continuity is verified by the performance of operational tests.

4.6 **Tests of duel tape transmitter.**

4.6.1 *Motor adjustment.—60 speed (AN/TGC-5( )X only).*—Using a standard stroboscopic tuning fork, operating at 180 vibrations per second, the motor speed shall be adjusted so that the transmitter operates at 368.1 operations per minute.

4.6.1.1 *Motor adjustment—100 speed (AN/TGC-5( )X only).*—Using the same tuning fork as in 4.6.1, the motor speed shall be adjusted so that the transmitter operated at 600 operation per minute.

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4.6.2 *Test of operation as a tape reader—60 speed.*—A test tape shall be inserted in each transmitter and the output connected to a five-wire teletypewriter, or equal. The transmitters shall then be pulsed by a 25 millisecond pulse at a pulse rate of 368.1 pulses per minute. The received intelligence from each transmitter shall be checked with the test tapes to insure that no errors have been made.

4.6.2.1 *Test of operation as a tape reader—100 speed.*—The test of 4.6.2 shall be repeated with a pulse rate of 600 pulses per minute.

4.7 **Rectifier test.**—Rectifier shall be tested as follows:

4.7.1 *Input test.*—Input voltage shall be varied over the range of 115±10 volts a. c. and the frequency from 50 to 60 cycles. It shall be possible, by adjusting frequency and voltage taps, to achieve 120±3 volts d. c. under any combination of the above conditions.

4.7.2 *Regulation.*—With a constant input voltage, the output voltage shall not vary by more than 12 volts when the current is varied from the minimum required during operation of the equipment to the maximum required to operate the equipment.

4.7.3 *Ripple voltage.*—The a. c. ripple voltage in the d. c. output when operated from input voltages and frequencies of 4.7.1 shall not exceed 0.5 volt at any point in the output d. c. current range.

4.8 **Typing reperforators.**—All typing reperforators shall conform to the requirements of Specification MIL-R-11985.

4.9 **Operational tests.**—The following operational tests shall be met.

4.9.1 *Break lock-normal single operation.*—The equipment shall be set up for operation at 60-speed with the normal-split switch on normal. The duplex-single switch shall be positioned to single and the polar-neutral switch shall be positioned to neutral 60 ma. A tape shall be inserted in a transmitter and the start lever raised. As soon as the message starts transmitting, open the signal line for a period of 300 milliseconds. The transmitter shall stop sending and its motor as well as the motor of the monitor reperforator shall be shut off. The receiving reperforator shall remain locked out with its motor left running. The break lamps shall be turned on. If the break release button is pushed the break lamps shall remain on and the motors shall not start unless the line is closed. The same test shall be performed for neutral 20 ma.

4.9.1.1 *Break lock-normal duplex operation.*—The equipment shall be set up as in 4.9.1, this time with the duplex single switch positioned to duplex operation. Again a break shall be introduced into the line. The same results shall be obtained. The same test shall be performed for neutral 20 ma operation and for polar operation. In polar operation, a break is construed to mean either spacing current or absence of current.

4.9.2 *Break lock-split single operation.*—The equipment shall be set up for operation at 60-speed with the normal-split switch on split. The duplex single switch shall be set on single and the polar-neutral switch shall be set on neutral 60 ma. A message tape shall be placed in each transmitter and the start lever of one raised. As soon as the message tape starts transmitting, introduce a 300 millisecond break into the line. The transmitter shall stop sending and its motor shall be shut off. The receiving reperforator associated with that line shall remain locked out with its motor left running. The break lamp shall be turned on. If the break release key for this line is pressed, the transmitter motor shall not start unless the line is closed. The same test shall be performed for neutral 20 ma operation. Repeat the above test for neutral 60 ma operation and with a break registered on line one, raise the start lever of the transmitter associated with line two. The transmitter motor shall start and transmitter No. 2 shall start to send whereas transmitter number one will not. Next introduce a break on line two. The transmitter shall stop sending and the transmitter motor shall stop. The break light associated with line two shall be lighted. With line one closed press the break release button associated with line one. The transmitter motor shall start and transmitter number one shall send. Transmitter number two shall not send and the break light shall remain on. Repeat above test for neutral 20 ma operation.

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4.9.2.1 *Break lock-split duplex operation.*—The test of 4.9.2 shall be repeated with the duplex-single switch set for duplex operation. Transmitter operational results shall be identical except that receiving reperforators are not locked out. Include test for polar operation.

4.9.3 *Motor control time delay test.*—Set up equipment for normal single neutral 60 ma operation with a monitoring page printer, or equal, in the line. Place a message tape in a transmitter and raise the start lever. The transmitter and monitor reperforator motor shall start and the transmitter will start to send. Inspect the copy at the page printer and at the monitor reperforator for any errors.

4.9.4 *Alternate transmitter operation.*—Set up equipment for 60-speed normal single neutral 60 ma operation and place message tapes in each transmitter. Raise the start lever of transmitter number one. As soon as transmitter starts, raise the start lever for transmitter number two. When the tape in transmitter number one clears the end of tape sensing pin, transmission of numbering information shall be sent from transmitter number two. Place another tape in transmitter number one and raise the start lever. As soon as the tape has cleared transmitter number two, numbering information shall be sent from transmitter number one. During the interval between message intelligence neither the transmitter nor monitor reperforator motors shall change speed. The test shall be repeated for normal duplex 60 ma operation.

4.9.4.1 Repeat tests of 4.9.4 with equipment arranged for 100-speed.

4.9.5 *Reperforator lock-out—normal operation.*—The equipment shall be set up for 60-speed normal single neutral 60 ma operation. A message tape shall be placed in a transmitter and the start lever raised. When sending starts, the monitor reperforator shall make a copy of the sent message and the receiving reperforator shall be locked out with its motor running. Signals shall be transmitted to the equipment and the receiving reperforator shall make a copy of the sent message while the monitor reperforator shall be locked out and its motor shut off.

4.9.5.1 *Reperforator lock-out—split operation.*—The equipment shall be set up for 60-speed split single neutral 60 ma operation and message tapes placed in the transmitter. Both start levers shall be raised and during transmission, both receiving reperforators shall be locked out. Messages shall next be sent on each line to the equipment and the receiving reperforators shall make a copy of the respective messages.

4.9.6 *Generation of blank signal test—60-speed.*—The equipment shall be set up for 60-speed normal single neutral 60 ma operation and message tapes placed in the transmitter. Both transmitters shall be set up for the generation of twenty blank characters. Both start levers shall be raised. At the end of transmission, the monitor tape shall be inspected to insure that twenty blanks plus or minus one have been transmitted.

4.9.6.1 *Generation of blank signal test—100-speed.*—The same test shall be repeated with the transmitters adjusted for 100-speed operation.

4.9.7 *Local test circuit test—60-speed.*—The equipment shall be set up for 60-speed local test neutral 60 ma operation. A message tape shall be placed in each tape transmitter. The start lever of transmitter number one shall be raised. As soon as transmitter starts, the start lever for transmitter number two shall be raised. Transmitter number one shall start sending and upon completion of message the second transmitter shall start sending with both reperforators copying the message. The same test shall be performed for neutral 20 ma operation and for polar operation. During this test external teletypewriter apparatus connected to the equipment shall not be disturbed when switching from any mode of operation into local test condition.

4.9.7.1 *Local test circuit test—100-speed.*—The tests of 4.9.7 shall be repeated for 100-speed operation.

4.9.8 *Automatic numbering test—60-speed.*

4.9.8.1 *Normal single 60 ma operation.*—The equipment shall be set up and short message tapes placed in each transmitter. The message numbering reset button shall be operated to insure that the first number will be 001. Both start levers shall be raised. Five messages shall

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be set in this manner and the monitor tape checked to insure that the sequences of 3.10.7.7 have been transmitted.

4.9.8.2 *Split single 60 ma operation.*—The equipment shall be set up and message tapes placed in each transmitter. Receiving units shall be placed on outgoing line terminals for monitoring. Both message numbering reset buttons shall be operated to insure that the first numbers will be 001. Five short messages shall be sent from each transmitter and the sent copy checked as in 4.9.8.1.

4.9.8.3 *Test of disablement of automatic numbering.*—The test of 4.9.8.1 and 4.9.8.2 shall be repeated with the equipment adjusted so that automatic numbering is disabled. No numbering sequences shall be sent to the line.

4.9.8.4 *Automatic numbering test—100-speed.*—The tests of 4.9.8.1 and 4.9.8.2 shall be repeated with the teletypewriter equipment adjusted for 100-speed operation.

4.9.9 *Radio interference suppression test.*—The equipment shall be tested for compliance with 3.14.

4.10 **Service condition test.**—The following service condition test shall be performed on one preproduction sample at a Government laboratory, as hereinafter specified. During the service conditions test, the equipment will be operated continuously or intermittently, as the specification requires, and performance data recorded. Between tests, the equipment shall be checked for such defects as low dielectric strength, loss of electrical continuity, loose parts, excessive corrosion, breakage, etc. Performance checks shall be made on the equipment at the conclusion of each service condition test. These performance checks shall be sufficiently adequate in number and scope, to uncover any defects resulting from the preceding service conditions test. Any evidence of degradation of performance below limits specified herein shall be cause for rejection.

4.10.1 *Low temperature test.*—The equipment shall be maintained for 72 hours in a chamber at an ambient temperature of −80° F. The temperature shall then be raised to +32° F. and stabilized for 1 hour. It shall then be subjected to the appropriate electrical tests for performance measurement.

4.10.2 *High temperature test.*—The equipment shall be maintained for 72 hours in a chamber at an ambient temperature of +160° F. The temperature shall then be lowered to +132° F. and stabilized for 1 hour. It shall then be subjected to the appropriate electrical tests for performance measurement.

4.10.3 *Altitude test.*—The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure, and humidity, and performance shall be measured. The pressure shall then be reduced to 20 inches of mercury and stabilized for 2 hours. The performance of the equipment shall again be measured. The pressure shall then be lowered to 5½ inches of mercury and the chamber maintained at this pressure for another 2-hour period. The pressure shall then be increased to 29.9 inches of mercury and the performance measured again. There shall be no degradation of performance in the second and third measurements from those established in the first test.

4.10.4 *Moisture-resistance test.*

4.10.4.1 *Test conditions.*

(a) Do not move components from the humidity chamber for measurements.

(b) Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.10.4.2 *Test procedure.*—The components will be tested as follows:

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Condition at 77° ± 5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Measure the performance of the sample and readjust if necessary to meet specification requirements.

(d) Subject the sample to continuous cycling for a total of 5 of the 48-hour cycles on Drawing SC-D-15914 and measure the performance in accordance with specification requirements, during the periods indicated on that drawing.

4.10.4.3 *Performance.*—At the completion of moisture-resistance test, repeat the inspection and tests of 4.10.4.2 (c). There shall be no mechanical deterioration, as evidenced by corrosion of metal parts or binding of rotating parts, and there shall be no degradation in performance from the specification requirements.

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4.10.5 *Bounce test, three-hour.*—The equipment, as packaged for shipment, shall be placed on the table of the Package Tester as made by the L. A. B. Corporation, Summit, N. J., or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 revolutions per minute (r. p. m.), ±1 percent, for a total of 3 hours. During each ½-hour period of the test, the packaged equipment shall rest on a different face. The equipment shall have no degradation of performance but surface abrasion is permissible.

4.10.6 *Vibration test.*—The equipment shall be tested for resonant frequencies below 55 cycles per second, to determine compliance with 3.4.2 (e) as follows:

(a) The equipment shall be fastened in its normal mounting position on a vibration table and that can be controlled within 10 percent of the specified amplitude. Equipment shock mounts, if any, shall be blocked. The vibration table shall provide approximately sinusoidal vibration. (See 4.10.6 (d).)

(b) The equipment shall be vibrated successively in three mutually perpendicular directions that are parallel respectively to the edges of the equipment, over a frequency range of 10 to 55 cycles per second, in 1-cycle-per-second steps. The total excursion shall be constant at 1/64 inch.

(c) Mechanical resonance, if any, of the complete structure, of subassemblies, and of component parts shall be determined visually by means of a Strobotac, as made by the General Radio Corporation, Cambridge, Mass., or equal, or by energizing the equipment and detecting mechanical vibration through electrical output indications.

(d) If it is not practical to determine completely any mechanical resonances on the assembled equipment unit because the unit weighs more than 250 pounds or because of its compactness or the inaccessibility of its subassemblies or component parts, the unit shall be broken down into logical subassemblies for the above vibration test. Such subassemblies shall be secured to the vibration table in the same manner as they are secured in the equipment unit.

4.10.7 *Life test.*—The equipment, with the motor drive and pinion gear used for 600 o. p. m. installed, shall be connected to a 20 milliampere neutral type circuit and to a Distortion Test Set TS-383/GG, or equal, to receive a repeated message for a total of 2,000 hours operation, following routine maintenance and adjustment schedule after each 250 hours of operation. The unit shall be tested at the maintenance intervals and at conclusion of the life test to determine compliance with 3.4.1. Degradation of performance shall not be greater than 5 percent of the operational requirements of this specification. Failure of any component by breakage or wear sufficient to cause inoperation, shall not be considered failure of the equipment as a whole.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *For domestic shipment.*—Components of Teletypewriter Sets AN/TGC-5( ) and AN/TGC-5( )X shall be packaged in accordance with method III of Specification MIL-P-116 (see 6.2.)

5.1.2 *For overseas shipment.*—Components of Teletypewriter Sets AN/TGC-5 ( ) and AN/TGC-5 ( )X shall be preserved and packaged in accordance with applicable methods prescribed in Specification MIL-P-116. (See 6.2.)

5.1.3 *Pilot pack.*—Upon determination of processes of cleaning and preservation, and the method of packaging to be used in the unit protection of the components, pilot packs shall be prepared by the contractor and, prior to quantity production, tested in accordance with the applicable provisions of Specification MIL-P-116, except that final approval and acceptance shall be made by a packaging representative of the procuring activity.

5.2 Packing.

5.2.1 *For domestic shipment.*—Components of Teletypewriter Sets AN/TGC-5( ) and AN/TGC-5( )X, packaged as specified in

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paragraph 5.1 shall be packed in suitable domestic shipping containers conforming to the provisions of Specifications NN-B-621, NN-B-631 or LLL-B-631, as applicable (see 6.2.)

5.2.2 *For overseas shipment.*—Components of Teletypewriter Sets AN/TGC-5( ) and AN/TGC-5( )X, packaged as specified in paragraph 5.1 shall be packed in suitable overseas shipping containers conforming to the provisions of Specifications JAN-P-104, JAN-P-105, or JAN-P-106, as applicable (see 6.2.)

5.3 **Marking.**—Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

6. NOTES

6.1 **Intended use.**—Teletypewriter Sets AN/TGC-5( ) and AN/TGC-5( )X provides the necessary facilities for transmission and reception of teletype messages in fixed station installations.

6.2 **Ordering data.**—Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Whether domestic or overseas shipment is required (see 5.1.1, 5.1.2, 5.2.1 and 5.2.2.)
(c) Samples required (see 3.2.1.)
(d) Marking and shipping of samples.
(e) Place of final inspection (see 4.1.)
(f) Technical literature required (see 3.17.)
(g) Quantity of tools and running spare parts required (see 3.17.)
(h) Facsimiles required (see 3.13.)

6.3 **Contractual requirements.**—It is recommended that the following be included in the contract:

6.3.1 *Additional testing.*—It should be understood that additional inspection and nondestructive testing of equipment may be performed by the Government inspector when considered necessary to determine compliance with this specification and other applicable documents. The Government inspector, therefore, may withdraw materials or parts temporarily from production for such inspection and testing, performed either at a Government laboratory or the contractor's plant.

6.3.2 *Waiver of preproduction samples.*—When items for which preproduction samples are required are currently in production on a Government order, the contractor may apply to the contracting officer for waiver of preproduction samples on the new order. Any request for waiver should include a statement indicating that the specified samples have been submitted and approved on the current contract; production of the item under the new order will be concurrent with, or follow without interruption, production on the current order; and current production is in strict compliance with specified requirements for the new equipment, or describing any deviations authorized on the current order. Waiver of preproduction samples may be contingent on renegotiation of the contract to reflect the saving in cost to the contractor.

6.4 **Nomenclature.**—The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example, AN/TGC-5W. As soon as possible after the award of contract, the contractor should apply to the contracting officer for such information.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no repsonsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodians:
Army: Signal Corps
Air Force.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD. FORM 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE S-0279

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